Hardware Test: Step-by-Step Guide

WebRTC address leakage and client isolation on real Wi-Fi
Companion to the paper in the `computernetpaperwork` folder

1. What this test is

The paper's experiment so far ran on an emulated network inside the laptop, with desktop browsers. This test repeats the two most important measurements on real equipment: a real Wi-Fi router and real phones. It answers two questions.

- What address does each phone browser give away during a WebRTC connection?
- When the router's "client isolation" is switched on, can two devices on the same Wi-Fi still connect to each other?

The test only watches how ordinary browsers behave. It does not attack anything.

Rule: use only a router you own or are allowed to configure, and only your own or your friends' devices with their agreement. Do not run this on the college network unless the IT department has given written permission.

2. What you need

- A Wi-Fi router whose settings you can open (a home router is fine).
- This laptop, with Python installed (it already is).
- An Ethernet (LAN) cable to connect the laptop to the router. This matters: when isolation is on, phones cannot reach a laptop that is on Wi-Fi, but they can normally reach one that is plugged in by cable.
- An Android phone with Chrome. Firefox for Android as well, if you can install it.
- An iPhone with Safari (borrow one if needed). iPhone results are new: no published study we used has them.
- Internet access through the router. The test page uses a public Google server to find each device's outside address.
- About one to two hours.

3. One-time setup

1. Plug the laptop into one of the router's LAN ports with the cable. Turn the laptop's Wi-Fi off so it uses only the cable.
2. Open the folder `hardware_test`, which is inside `computernetpaperwork` on the Desktop. Click the address bar of the folder window, type `cmd` and press Enter. A black command window opens.
3. Type `python server.py` and press Enter. It prints one or more web addresses, for example

   `http://192.168.1.5:8000/`

   Leave this window open for the whole test.

4. If Windows shows a firewall question, tick *Private networks* and choose *Allow*.

5. Connect each phone to the router's Wi-Fi. On each phone, open the browser and type the printed address exactly, including `:8000`. A page titled "WebRTC path test" should appear.

If the page does not open: check that the phone is on the same router, try the other printed address (use the one starting with `192.168` or `10.`), and check the firewall step.

4. How to run one test

1. On both devices, set the same **Room number** (for example 1).

2. On each device, fill **This device and browser** (for example "Android Chrome") and **Network setup** (for example "isolation OFF").

3. On the first device press **Start as A**.

4. Within two minutes, on the second device press **Join as B**.

5. Wait up to 30 seconds. Each device shows a line beginning **Result:** and a line saying what the device advertised.

6. Copy both into the record sheet (Section 6). The laptop also saves every result automatically in `results.jsonl`.

7. Reload the page on both devices before the next test.

The result is one of:

- **DIRECT (device to device)**: the two devices talked to each other over the local network.
- **DIRECT via reflexive address**: they connected using their outside-visible addresses.
- **NO CONNECTION**: they could not connect. This page offers no relay server, so a blocked direct path shows up as no connection.

5. Test plan

Run every device pair under every router setup, three times each.

Setup	Router setting	How to set it
S1	Client isolation **OFF**	The normal state of most home routers. Check that "AP isolation", "client isolation" or "wireless isolation" is off in the wireless settings.
S2	Client isolation **ON**	Turn the same setting on and save. Phones may need to reconnect to the Wi-Fi. If the router has no such setting, note that and skip S2.
S3 (optional)	Guest network	Put both phones on the router's guest Wi-Fi, which usually isolates clients. If the test page stops opening on the guest network, write that down and stop S3; it means guests cannot reach the wired side.

Pair	Device A	Device B
P1	Android, Chrome	iPhone, Safari
P2	Android, Firefox	iPhone, Safari
P3	Android, Chrome	second Android phone, Chrome (a friend's)
P4 (optional)	Any phone	The laptop's Chrome (laptop is on cable, so this is wireless-to-wired; record it separately)

Minimum useful set: S1 and S2, pairs P1 and P3, three times each. That is 12 tests.

IPv6 check. On one phone, on the test Wi-Fi, look at the "This device advertised" line. If it shows an address containing colons (for example `2401:...`), your connection has IPv6. Write "IPv6 yes" on the sheet. This matters because the paper found that global IPv6 addresses are given away in every session.

6. Record sheet

Write these once: date; router brand, model and firmware version; each phone's model and operating-system version; each browser's version (in the browser's *About* screen).

Then one row per test:

Setup	Pair	Run	Result line	A advertised	B advertised
S1	P1	1			
S1	P1	2			
S1	P1	3			
S2	P1	1			
S2	P1	2			
S2	P1	3			
S1	P3	1			
S1	P3	2			
S1	P3	3			
S2	P3	1			
S2	P3	2			
S2	P3	3			

7. What the results would mean

- **A phone shows "RAW local address"**: that browser gives away the device's real Wi-Fi address. The 2025 study reported this for Android Chrome and Firefox; your result would confirm or contradict it on current versions.
- **A phone shows only a "hidden .local name"**: that browser hides the local address.
- **S1 gives DIRECT**: expected. Without isolation, devices on one Wi-Fi can reach each other.

- **S2 gives NO CONNECTION**: the router's isolation really blocks device-to-device traffic. It also shows that apps without a relay server stop working between two people on the same isolated Wi-Fi.
- **S2 still gives DIRECT**: the most interesting outcome. It means isolation on this router does not stop two devices from reaching each other, which is what the emulation found when the gateway forwards between clients.
- **An address with colons appears**: the device has IPv6 and is giving that address away.

Whatever the outcome, report it as it is. A result that disagrees with the paper is still a result.

8. When you finish

1. Press Ctrl+C in the command window to stop the server.
2. Keep `results.jsonl` and a photo or typed copy of the record sheet in the `hardware_test` folder.
3. Turn the router's isolation setting back to how it was.
4. Ask for the results to be added to the paper as a "Hardware validation" section.

Limits to state in the paper. One home router and a few phones are not a campus network. The test has no relay server, uses a public STUN server, and each case is run only three times. It strengthens the paper; it does not make the results general.